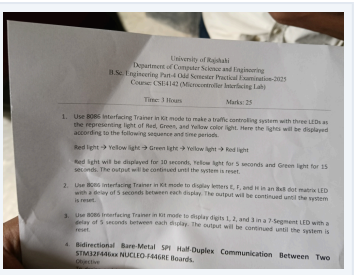
8086 Traffic Control System Breakdown

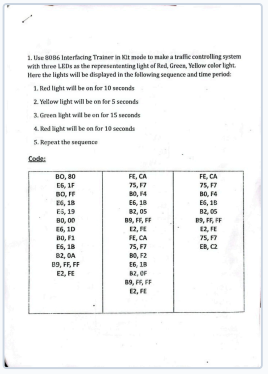


This document details the line-by-line execution of the 8086 machine code designed to simulate a traffic control system. The sequence cycles continuously through Red (10s), Yellow (5s), Green (15s), and Yellow (5s) states utilizing the 8255 Programmable Peripheral Interface (PPI).

Hex Code	Assembly Instruction	Action / Technical Explanation
1. HARDWARE INITIALIZATION		
`B0, 80`	`MOV AL, 80H`	Load control word `80H` (10000000b) to configure all 8255 PPI ports as Output (Mode 0).
`E6, 1F`	`OUT 1FH, AL`	Send the configuration control word to the 8255 Control Register (Port address `1FH`).
`B0, FF`	`MOV AL, FFH`	Load `FFH` (all 1s) to turn OFF all LEDs, assuming an active-low LED configuration.
`E6, 1B`	`OUT 1BH, AL`	Output `FFH` to Port B (address `1BH`), clearing any previously lit LEDs.
`E6, 19`	`OUT 19H, AL`	Output `FFH` to Port A (address `19H`), ensuring all output pins are reset.
`B0, 00`	`MOV AL, 00H`	Clear the AL register by loading `00H`.
`E6, 1D`	`OUT 1DH, AL`	Output `00H` to Port C (address `1DH`), finishing the kit reset sequence.
2. STATE 1: RED LIGHT (10 SECONDS)		
`B0, F1`	`MOV AL, F1H`	Load hex pattern `F1H` to activate the Red LED.
`E6, 1B`	`OUT 1BH, AL`	Send the Red LED pattern to Port B to physically turn on the light.
`B2, 0A`	`MOV DL, 0AH`	Load the outer delay loop multiplier into DL. `0AH` (decimal 10) sets a 10-second wait.
`B9, FF, FF`	`MOV CX, FFFFH`	Initialize the inner loop counter CX to its maximum 16-bit value (65535).
`E2, FE`	`LOOP FE`	Tight inner loop: decrements CX and jumps back to itself until CX reaches 0. Creates the base unit of delay time.
`FE, CA`	`DEC DL`	Decrement the outer loop multiplier (DL) after one full cycle of the inner loop finishes.
`75, F7`	`JNZ F7`	Jump back -9 bytes to `MOV CX, FFFFH` if DL > 0. Repeats the inner loop exactly 10 times.

Delay Loop Architecture

The required time delays (5s, 10s, 15s) are generated using a nested loop structure. The CX register handles a tight inner loop (counting down from FFFFH), while the DL register tracks the requested outer loop seconds, governed by the JNZ instruction.



Hex Code	Assembly Instruction	Action / Technical Explanation
3. STATE 2: YELLOW LIGHT (5 SECONDS)		
`B0, F4`	`MOV AL, F4H`	Load hex pattern `F4H` to activate the Yellow LED.
`E6, 1B`	`OUT 1BH, AL`	Send the Yellow LED pattern to Port B.
`B2, 05`	`MOV DL, 05H`	Load the outer delay loop multiplier into DL. `05H` (decimal 5) sets a 5-second wait.
`B9, FF, FF`	`MOV CX, FFFFH`	Reset the inner loop counter CX to `FFFFH` (65535).
`E2, FE`	`LOOP FE`	Execute inner base delay loop until CX becomes 0.
`FE, CA`	`DEC DL`	Decrement the outer loop multiplier (DL).
`75, F7`	`JNZ F7`	Jump back to `MOV CX, FFFFH` if DL > 0. Completes exactly 5 delay cycles.
4. STATE 3: GREEN LIGHT (15 SECONDS)		
`B0, F2`	`MOV AL, F2H`	Load hex pattern `F2H` to activate the Green LED.
`E6, 1B`	`OUT 1BH, AL`	Send the Green LED pattern to Port B.
`B2, 0F`	`MOV DL, 0FH`	Load the outer delay loop multiplier into DL. `0FH` (decimal 15) sets a 15-second wait.
`B9, FF, FF`	`MOV CX, FFFFH`	Reset the inner loop counter CX to `FFFFH`.
`E2, FE`	`LOOP FE`	Execute inner base delay loop until CX becomes 0.
`FE, CA`	`DEC DL`	Decrement the outer loop multiplier (DL).
`75, F7`	`JNZ F7`	Jump back to `MOV CX, FFFFH` if DL > 0. Completes exactly 15 delay cycles.

Hex Code	Assembly Instruction	Action / Technical Explanation
5. STATE 4: YELLOW LIGHT RETURN (5 SECONDS)		
`B0, F4`	`MOV AL, F4H`	Load hex pattern `F4H` again to reactivate the Yellow LED.
`E6, 1B`	`OUT 1BH, AL`	Send the Yellow LED pattern to Port B.
`B2, 05`	`MOV DL, 05H`	Load the outer delay loop multiplier into DL for the final 5-second wait.
`B9, FF, FF`	`MOV CX, FFFFH`	Reset the inner loop counter CX to `FFFFH`.
`E2, FE`	`LOOP FE`	Execute inner base delay loop until CX becomes 0.
`FE, CA`	`DEC DL`	Decrement the 5-second outer loop multiplier (DL).
`75, F7`	`JNZ F7`	Jump back to reset CX until the 5-second duration is completed.
6. LOOP CONTINUOUSLY		
`EB, C2`	`JMP C2`	Unconditional relative jump backward. `C2` points the Instruction Pointer back to the exact memory address of `B0, F1`, restarting the Red Light sequence infinitely.

Execution Summary

This 8086 kit assembly routine successfully fulfills the objective of creating a continuous 4-state traffic light system. By carefully managing the AL register for Port B data transmission, specific LEDs are toggled on and off utilizing the hex masks (F1H, F4H, and F2H). Timing precision is achieved by nesting a hardware loop (LOOP FE, tracking CX) inside a software decrement check (tracking DL). The terminal JMP instruction ensures the system remains active until a manual kit reset occurs.